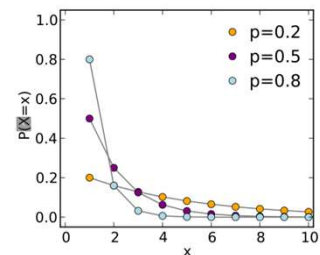
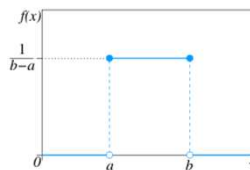
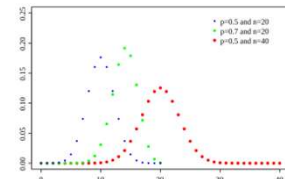
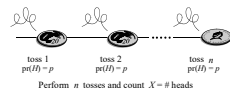
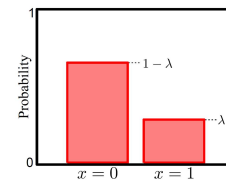
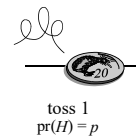


Agenda

- Last time
 - Some distributions used over and over again
 - Hypergeometric
 - Continuous distribution
- Today
 - PRNG
 - Uniform distribution
 - Random subset



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Hypergeometric

$$P(X = i) = \frac{\binom{N}{i} \binom{M}{n-i}}{\binom{N+M}{n}}$$

- If we sample n balls at random from a finite population without replacement, and count X , the number of black balls
- Consider X_i which has value 1 if the i th trial produces a black ball and 0 otherwise
 - Let $T = N + M$
 - Clearly, $P(X_1 = 1) = \frac{N}{T}$
 - Assume $P(X_k = 1) = \frac{N}{T}$
- Although the X_i 's are **not independent**, every position in the sequence has the same marginal probability

$$\begin{aligned}
 P(X_{k+1} = 1) &= P(X_{k+1} = 1 \mid X_k = 1)P(X_k = 1) \\
 &\quad + P(X_{k+1} = 1 \mid X_k = 0)P(X_k = 0) \\
 &= \frac{N-1}{T-1} \frac{N}{T} + \frac{N}{T-1} \frac{M}{T} \\
 &= \frac{N(N-1) + NM}{T(T-1)} \\
 &= \frac{N(T-1)}{T(T-1)} \\
 &= \frac{N}{T}
 \end{aligned}$$

N black balls
M white balls
Choose n balls
How many black?

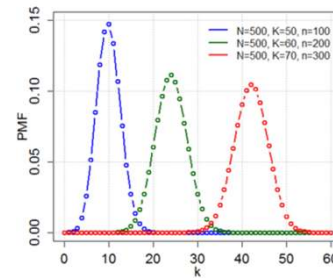


3

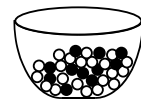
Hypergeometric

$$P(X = i) = \frac{\binom{N}{i} \binom{M}{n-i}}{\binom{N+M}{n}}$$

- If we sample n balls at random from a finite population and count X , the number of black balls
- How do we choose “at random” 100 people from a group of 200?



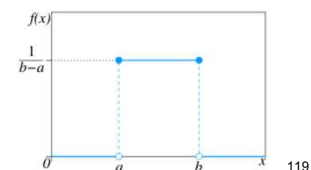
N black balls
M white balls
Choose n balls
How many black?



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Uniform distribution

- Uniform random variables, especially over the $[0,1]$ interval are very important, in developing programs to draw samples from other distributions
- That's the theory
 - In practice, we assume that we have access to a generator that can generate a canonical random variable
 - The random number generator is actually a pseudo-random number generated



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Pseudorandom Numbers

- Concept of a seed
- For any initial seed, subsequent numbers appear to be random, cannot be predicted ...

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Demo

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Random Numbers: Produce integers

- `rand()` is often used

What's **Wrong** With This Code?

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
    srand(time(NULL));
    for (int i = 0; i < 16; ++i) {
        printf("%d ", rand() % 100);
    }
    printf("\n");
}
```

Adapted from `rand()` considered harmful (Stephan T. Lavavej)

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Random Numbers (API)

What's **Right** With This Code?

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
    srand(time(NULL));
    for (int i = 0; i < 16; ++i) {
        printf("%d ", rand() % 100);
    }
    printf("\n");
}
```

All required headers are included!

All included headers are required!

Headers are sorted!

One True Brace Style!

%d is correct for int!

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Random Numbers (API)

- `rand()` is often used

What's **Wrong** With This Code?

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
    srand(time(NULL));
    for (int i = 0; i < 16; ++i) {
        printf("%d ", rand() % 100);
    }
    printf("\n");
}
```

Frequency: 1 Hz!

32-bit seed!

warning C4244:
'argument' :
conversion from
'time_t' to
'unsigned int',
possible loss of
data

Range: [0, 32767]

Non-uniform distribution!

Linear congruential → low quality!

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Random Numbers (API)

- Assume `rand()` is uniform in [0, 32767]

// [0, 99] src → [0, 99] dest

// [100, 199] src → [0, 99] dest

// ...

// [32700, 32767] src → [0, 67] dest

- The output is not uniform

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Hello random world

- Mersenne twister
 - Not cryptographically secure
- Automatic seed

```
import random
//random.seed(42)
n = int(input())
a = []
for i in range(n):
    a.append(random.randint(1, 100))
for i in range(n - 1):
    min_index = i
    for j in range(i + 1, n):
        if a[j] < a[min_index]:
            min_index = j
    a[i], a[min_index] = a[min_index], a[i]
for x in a:
    print(x, end=" ")
print()
```

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Hello, "Random" World!

```
#include <iostream>
#include <random>
int main() {
    std::mt19937 mt(1729);
    std::uniform_int_distribution<int> dist(0, 99);
    for (int i = 0; i < 16; ++i) {
        std::cout << dist(mt) << " ";
    }
    std::cout << std::endl;
}
```

Engine: [0, 2³²)

Deterministic 32-bit seed

Distribution: [0, 99]

Note: [inclusive, inclusive]

- Engine: fast, seedable, reproducible
- Obsessive implementors (perfect uniformity), deterministic but may vary across platforms

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Pseudorandom Numbers

- Concept of a seed
- For any initial seed, subsequent numbers appear to be random, cannot be predicted ...
- ... until we reach the period after which the numbers reappear in the same order so
 - We need a long period
 - Need efficient pseudo random numbers generator (prng) (should be as cheap as anything)

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Why uniform?

- Uniform random variables, especially over the $[0,1]$ interval are very important, in developing programs to draw samples from other distributions
- *We assume the existence of a canonical random number generator*

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Generating a random subset

- Choosing a population for a medical trial
- Generate a random subset B_k of size k from a set $A = a_1, a_2, \dots, a_n$ of size n
 - All the $\binom{n}{k}$ subsets must be equally likely
- Consider another problem
 - Generate a discrete random variable X having the pmf

$$P\{X = x_j\} = p_j, \quad j = 0, 1, \dots, \sum_j p_j = 1$$

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Generating a discrete pmf

- Generate a discrete random variable X having the pmf
- Example: 0.2, .15, .25, 0.4
- Algorithm
 - Generate a sequence of random numbers U that is uniformly distributed over $[0, 1]$

$$P\{X = x_j\} = p_j, \quad j = 0, 1, \dots, \sum_j p_j = 1$$

$$X = \begin{cases} x_0 & \text{If } U < p_0 \\ x_1 & \text{If } p_0 \leq U < p_0 + p_1 \\ \vdots & \vdots \\ x_j & \text{If } \sum_{i=0}^{j-1} p_i \leq U < \sum_{i=0}^j p_i \\ \vdots & \vdots \end{cases}$$

$$0 < a < b < 1 \longrightarrow P\{a \leq U < b\} = b - a$$

$$P\{X = x_j\} = P\left\{\sum_{i=0}^{j-1} p_i \leq U < \sum_{i=0}^j p_i\right\} = p_j$$

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Generating a random subset

- Generate a random subset B_k of size k from a set $A = a_1, a_2, \dots, a_n$ of size n
 - All the $\binom{n}{k}$ subsets must be equally likely
- Define indicator variable

$$I_j = \begin{cases} 1 & \{a_j \in B_k\} \\ \text{else} & 0 \end{cases}$$

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Generating a random subset

- $P(I_1 = 1) = k/n$
 - Denominator all outcomes: $\binom{n}{k}$ ways to pick k objects out of n
 - Numerator (favourable outcomes)
 - Item 1 is picked, there is only one way of picking it, and
 - we have to pick $k-1$ items out of the remaining $n-1$ items i.e. $\binom{n-1}{k-1}$
 - Probability: favourable divided by total.
- If I_1 is 1, then having chosen I_1 , the remaining $k-1$ items must be chosen to be “equally probable”
 - $P(I_2 = 1 \mid I_1 = 1) = \frac{k-1}{n-1}$

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Generating a random subset

- $P(I_1 = 1) = k/n$
 - $P(I_2 = 1 \mid I_1 = 1) = \frac{k-1}{n-1}$
 - If I_1 is 0, $P(I_2 = 1 \mid I_1 = 0) = \frac{k}{(n-1)}$
- Therefore, $P(I_2 = 1) = \frac{(k-I_1)}{(n-1)}$
- Continuing,

$$P(I_j = 1 \mid I_1, \dots, I_{j-1}) = \frac{k - \sum_{i=1}^{j-1} I_i}{n - (j-1)}$$

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Generating a random subset

- Use an algorithm similar to the earlier algorithm

$$U_j \sim \text{Uniform}(0, 1)$$

- Generate at most n random numbers with the stated probabilities
- Choose item 1 if $U_1 < k/n$ else skip
- Choose item 2 if $U_2 < (k-1)/(n-1)$ else skip
- ...

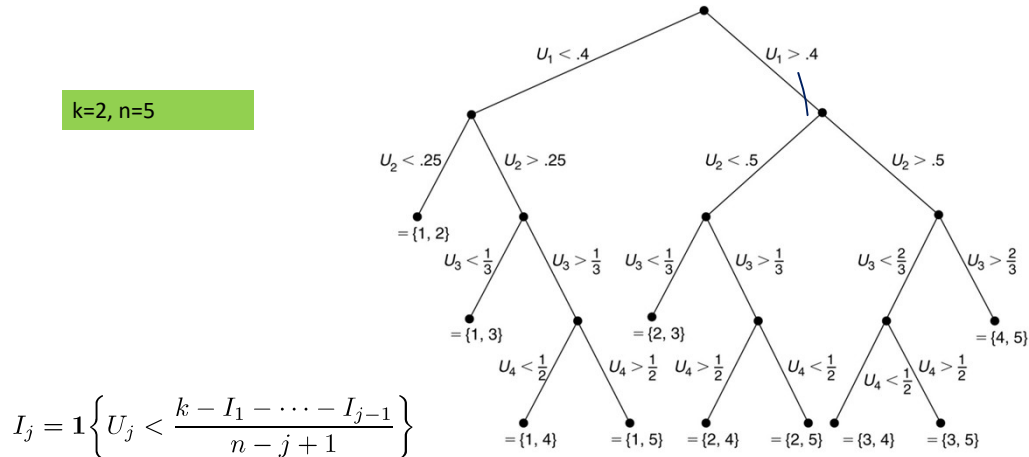
$$I_j = 1 \left\{ U_j < \frac{k - I_1 - \dots - I_{j-1}}{n - j + 1} \right\}$$

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Example

$k=2, n=5$



$$I_j = \mathbf{1} \left\{ U_j < \frac{k - I_1 - \dots - I_{j-1}}{n - j + 1} \right\}$$

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Generating a random subset

Algorithm stops

- Step j if $I_1 + I_2 + \dots + I_j = k$
 - Random subset B_k contains those indices whose indicator values are 1
- Or, if the number of unfilled entries in the random subset B_k equals the number of remaining elements in A

$$U_j \sim \text{Uniform}(0, 1)$$

$$I_j = \mathbf{1} \left\{ U_j < \frac{k - I_1 - \dots - I_{j-1}}{n - j + 1} \right\}$$

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